

RESEARCH INTEREST

My research interests lie in the intersection of Analysis, Geometry and Topology with possible application to Physics. I exploit tools from Index Theory, Combinatorial Topology, and Parametrised Stable Homotopy Theory to tackle challenging questions in Differential Geometry and Differential Topology with the main focus on the spaces and moduli spaces positive scalar curvature metrics, Riemannian metrics with the exceptional holonomy group G_2 and $\text{Spin}(7)$, and manifolds bundles with special tangential structures.

RESEARCH AND WORK EXPERIENCE

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| Postdoctoral Researcher , The University of Melbourne | Since 2024 |
| <ul style="list-style-type: none">– Conducting innovative and collaborative research on the global topology of moduli spaces of G_2 manifolds and its connection to theoretical physics– Volunteering in teaching by assisting in lectures and organising seminars; Supervising Master's students and Summer Research Scholars | |
| Postdoctoral Researcher , Albert-Ludwigs Universität Freiburg | 2022-2024 |
| <ul style="list-style-type: none">– Conducting innovative and collaborative research on the global topology of moduli spaces of G_2 manifolds within the Simon Collaboration "Special Holonomy in Geometry, Analysis and Physics"– Volunteering in teaching by assisting in lectures and organising seminars; Cosupervising Master's students | |
| Doctoral Researcher , Georg-August Universität Göttingen | 2017-2022 |
| <ul style="list-style-type: none">– Conducted collaborative research on positive scalar curvature and index theory within Project 09 "Diffeomorphisms and the Topology of Positive Scalar Curvature" of the SPP 2026 "Geometry at Infinity"– Conducted independent research on parametrised stable homotopy theory leading to publication– Presented complex research results on international conferences and in research seminars as invited speaker– Training of Master students and volunteering in teaching by organising and running a seminar on the Atiyah-Patodi-Singer Index Theorem (October 2020 - February 2021) | |
| Doctoral Researcher , Universität Augsburg | 2018-2020 |
| Research Stay, Mentor: Prof. Dr. Wolfgang Steimle | |
| <ul style="list-style-type: none">– Research Project: Isotopy Versus Concordance in Positive Scalar Curvature; subproject of PhD thesis aiming to study positive scalar curvature metrics beyond index theory.– Construction of a novel comparison space and a spectral sequence that measures the difference between the two concepts. (April 2018 - September 2020)– volunteered for teaching by organising and running a seminar for bachelor and master students on simplicial sets. (April - August 2020) | |
| Master Student , Georg-August Universität Göttingen | 2017 |
| <ul style="list-style-type: none">– Computing bordism groups of classifying spaces to answer a question of Edward Witten. | |

EDUCATION

- Ph.D in Mathematics**, Georg-August Universität Göttingen, Grade: 1.0 2017–2022
Supervisors: Prof. Dr. Thomas Schick and Prof. Dr. Wolfgang Steimle
– Thesis: “Concordances in Positive Scalar Curvature and Index Theory”
- M.Sc. in Mathematics**, Georg-August Universität Göttingen, Grade: 1.0 2014–2017
Supervisor: Prof. Dr. Thomas Schick
– Thesis: “Pin Bordism in Low Degrees of Classifying Spaces”
– Major Fields of Study: Differential Geometry and Algebraic Topology
- B.Sc. in Mathematics**, Georg-August Universität Göttingen, Grade: 1.9 2010–2014
Supervisor: Jun.-Prof. Dr. Henrik Seppänen
– Thesis: “Bertini’s Theorem”
– Major Fields of Study: Complex Analysis and Algebraic Geometry

ARTICLES

2. T. Hertl, “Moduli Spaces of Positive Curvature Metrics in Dimension Four and Beyond”, *Mathematische Zeitschrift*, Vol 312, Article 73 (2026)
1. T. Hertl, “Line Bundle Twists for Unitary Bordism are Ghosts”, *Algebraic & Geometric Topology* (**25**) (2025), 2053–2066.

PREPRINTS

1. D. Crowley, S. Goette, T. Hertl, “Path components of G_2 -moduli spaces may not be contractible”, <https://arxiv.org/pdf/2503.15829>, March 2025 submitted to *Journal of the European Mathematical Society (JEMS)* (under review)
2. T. Hertl, “Fibrewise Orbifold Resolutions with Applications to G_2 -moduli spaces”, June 2026, ArXiv:2606.21128
3. T. Hertl, I. Lim “Homotopical robustness of isometries on the Cayley plane”, August 2026, ArXiv:2608.10559
4. G. Frenck, T. Hertl, “The Concordance Space of Positive Scalar Curvature Metrics” (In preparation)
5. S. Cecchini, G. Frenck, T. Hertl, “ S^1 -stability for concordances” (In preparation)

SURVEYS

1. G. Frenck, T. Hertl, “The space of psc-concordances”, April 2026, to appear in *Geometry at Infinity - Final volume of the DFG Priority Programme 2026*

STUDENTS

- (3) Pinak Joshi, (co-supervision with Prof. Dr. Diarmuid Crowley) January 2026
Summer Research Scholar
- (2) Isla Wen Cong Lim, (co-supervision with Prof. Dr. Diarmuid Crowley) March 2025 - May 2026
Master Thesis, Title “Non-triviality of $B\text{Iso}(OP^2)_0 \rightarrow B\text{hAut}(OP^2)_0$ ”
- (1) Jan-Bennet Wiskant, (co-supervision with Prof. Dr. Thomas Schick) Sept. 2024 - July 2025
Master Thesis, Title: “Normal Pin^- -bordism of classifying spaces in low degrees”

TEACHING

- **Instructor**, The University of Melbourne Spring 2025
Lecture on Rational Homotopy Theory
- **Course Assistant**, The University of Melbourne Spring 2025
Complex Analysis
- **Co-organiser**, The University of Melbourne Spring 2024 - Fall 2026
(jointly with Prof. Dr. Diarmuid Crowley and Prof. Dr. Marcy Robertson)
Topology Seminar
- **Instructor**, Albert-Ludwigs Universität Freiburg Fall 2023
(jointly with Dr. Jonas Schnitzer)
Seminar on Operads in Algebra, Topology, and Physics
- **Instructor**, Albert-Ludwigs Universität Freiburg Spring 2023
Proseminar on Matrix Groups
- **Teaching Assistant**, Albert-Ludwigs Universität Freiburg Fall 2022
Analysis III
- **Teaching Assistant**, Georg-August Universität Göttingen Fall 2021
Ordinary Differential Equations (B.Mat.0030.Mp)
- **Instructor**, Georg-August Universität Göttingen and Universität Augsburg Fall 2020
(jointly with Prof. Dr. Thomas Schick and Dr. Artem Nepechiy)
Oberseminar on the Atiyah-Patodi-Singer Index Theorem (M.Math.4914)
- **Instructor**, Universität Augsburg Spring 2020
(jointly with Prof. Dr. Wolfgang Steimle)
Seminar on Simplicial Sets.
- **Tutor**, Universität Augsburg Fall 2019 –Spring 2020
Linear Algebra I and II
- **Tutor**, Georg-August Universität Göttingen Fall 2016 –Spring 2017
Algebraic Topology I and II
- **Tutor**, Georg-August Universität Göttingen Spring 2016
Functional Analysis
- **Tutor**, Georg-August Universität Göttingen Fall 2015
Mathematical Models of Physics
- **Tutor**, Georg-August Universität Göttingen Spring 2015
Analysis II
- **Tutor**, Georg-August Universität Göttingen Spring 2014
Mathematics for Physicist
- **Tutor**, Georg-August Universität Göttingen Fall 2013
Analysis III

SCIENTIFIC TALKS

- (31) “ G_2 Moduli Spaces May Have Holes” December 2025
Gauge-Theory and String Geometry, Australian National University (ANU), Canberra, Australia
- (30) “ G_2 Moduli Spaces May Have Holes” July 2025
Arbeitsgruppen-Seminar Ammann, University of Regensburg, Regensburg, Germany
- (29) “ G_2 Moduli Spaces May Have Holes” July 2025
Gauge Theory Seminar, Humbolt University, Berlin, Germany

- (28) “ G_2 Moduli Spaces May Have Holes” July 2025
Oberseminar Differentialgeometrie, University of Hamburg, Hamburg, Germany
- (27) “Concordances and Bordism of Positive Scalar Curvature Metrics” June 2025
Oberseminar Differentialgeometrie, University of Freiburg, Freiburg, Germany
- (26) “ G_2 Moduli Spaces May Have Holes” June 2025
Oberseminar Geometrie und Topologie, University of Göttingen, Göttingen, Germany
- (25) “ G_2 Moduli Spaces May Have Holes” May 2025
Differential Geometry Seminar, University of Adelaide, Adelaide, Australia
- (24) “The Holonomy Groups of Symmetric Spaces” May 2025
TOPS, University of Adelaide, Adelaide, Australia
- (23) “Topological Methods in G_2 -geometry” March 2025
AIM Workshop “New directions in G_2 geometry”, AIM, Los Angeles, USA
- (22) “Moduli Spaces of Positive Curvature Metrics” March 2025
Non-commutative geometry seminar, Texas A&M, College Station, USA
- (21) “Concordances in Positive Scalar Curvature and Index Theory” December 2024
Joint Meeting of the AMS, AustMS, and NZMS, Auckland, New Zealand
- (20) “On the Global Topology of G_2 Moduli Spaces” September 2024
The Geometry of Moduli Spaces in String Theory, MATRIX Institute, Creswick, Australia
- (19) “On the Global Topology of G_2 Moduli Spaces” June 2024
Workshop on topology, representation theory and higher structures, Isle of Skye
- (18) “On Moduli Spaces of G_2 Holonomy Metrics” May 2024
Final Simons Collaboration Meeting, Durham, North Carolina
- (17) “Moduli Spaces of Positive Curvature Metrics” May 2024
Oberseminar Differentialgeometrie, University of Augsburg
- (16) “Moduli Spaces of G_2 manifolds may have holes” April 2024
Bath Algebra, Geometry and Number Theory Seminar, University of Bath
- (15) “Moduli Spaces of G_2 Holonomy Metrics” April 2024
Bavarian Topology Meeting, University of Regensburg
- (14) “Moduli spaces of Positive Curvature Metrics” March 2024
Online Differential Geometry Seminar, YMSC Tsinghua University
- (13) “Moduli spaces of Positive Curvature Metrics” February 2024
Oberwolfach Workshop 2408 “Analysis, Geometry and Topology of Positive Scalar Curvature Metrics”
- (12) “On Homotopy Groups of (Observer) Moduli Spaces of G_2 Manifolds” February 2024
ULB geometry seminar, Université libre de Bruxelles
- (11) “Moduli Spaces in Riemannian Geometry” September 2023
Annual Meeting of the German Math Society, TU Ilmenau
- (10) “Moduli Spaces of Positive Curvature Metrics” September 2023
“Developments in Modern Mathematics”, University of Goettingen
- (9) “Concordances in Positive Scalar Curvature and Index Theory” July 2023
“Seminar Differentialgeometrie und Geometrische Analysis”, HU Berlin
- (8) “Line Bundle Twists for Unitary Bordism are Ghosts” April 2023
“Topology Seminar”, University of Melbourne
- (7) “Concordances in Positive Scalar Curvature and Index Theory” November 2022
“Oberseminar Topologie”, WWU Münster
- (6) “Positive Scalar Curvature from a Concordance Viewpoint” June 2022

“Dirac Operators in Topology, Geometry, and Representation Theory”, Cortona

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| (5) “Positive Scalar Curvature from a Concordance Viewpoint”
“Oberseminar Topologie”, University of Fribourg | May 2022 |
| (4) “Cubical Approximations for Positive Scalar Curvature Metrics”
“Arbeitsgruppenseminar”, University of Regensburg | December 2021 |
| (3) “Cubical Approximation for Positive Scalar Curvature Metrics”
DMV-ÖMV Jahrestagung 2021 - Sektion 09, Geometrie und Topologie | September 2021 |
| (2) “Cubical Models for Positive Scalar Curvature”
“Oberseminar Geometrie”, University of Augsburg | November 2020 |
| (1) “Cubical Approximation for Positive Scalar Curvature Metrics”
SPP 2026 - Geometry at Infinity, WWU Münster, Germany | April 2019 |

SERVICE WORK

Reviewer for zbMATH Open	since 2022
Volunteer Open Day University of Melbourne	18th Aug 2024 17th Aug 2025
Referee for <i>Geometry & Topology</i> and <i>Quarterly Journal of Mathematics</i>	
Grader for the lower-saxony round of the Math-Olympiad	2015-2024

PROGRAMMING LANGUAGES

- **Java:** Working experience
- **C:** Working experience
- **Python:** Familiar with
- **R:** Familiar with

LANGUAGES

- **German:** native
- **English:** C1-level (IELTS with score 8)
- **Spanish:** basic

ACADEMIC REFERENCES

Prof. Dr. Thomas Schick
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Prof. Dr. Sebastian Goette
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